



TOMAS CLAUDIO COLLEGES

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DESIGNING AN ERGONOMIC DIGITAL SYSTEM

CS 223 FINAL EXAMINATION



PRESENTED BY: ACAIN, PRINCESS B.

COURSE/YEAR/BLOCK: BSCS 2B

SUBJECT: CS 223

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ACTIVITY 1: SYSTEM PROPOSAL

Chrysopoeia: An Alchemical Interface for Multi-Tiered Asset Management and Universal Financial Transmutation

Problem Statement

Financial management is frequently perceived as a clinical, intimidating, and anxiety-inducing chore. My research indicates that traditional banking interfaces rely on high-alert colors (such as aggressive reds and harsh whites) and dense, cluttered spreadsheets that create a high "cognitive load."

This visual stress often leads to financial avoidance, where users ignore their records to avoid the associated stress. Users, particularly those already under economic pressure or time constraints, require a digital environment that prioritizes emotional comfort and transforms data entry into a focused, aesthetic ritual rather than a stressful audit.

Target Users

The system is engineered for a diverse, universal user base:

- **Time-Poor Professionals:** Individuals like Elena Vance who require a high-level "vibe-check" of their net worth in under five seconds without digging through menus.
- **Daily Wage Earners:** Users who need to monitor micro-cash flows and daily necessities without being overwhelmed by complex corporate jargon.
- **The Financially Curious:** Beginners seeking an artistic and non-intimidating gateway to financial literacy, allowing them to track growth without fear.

Purpose of the Application

The primary objective of **Chrysopoeia** is to democratize wealth management by lowering the emotional barriers to financial tracking. By utilizing the historical and symbolic theme of alchemy, the application transmutes the "lead" of financial stress into the "gold" of disciplined asset management. I intend to provide a digital sanctuary where users can monitor their assets with clarity, artistic inspiration, and ergonomic ease, ensuring that the act of managing money becomes a sustainable, daily habit.

Expected Features

- **The Initiation Portal (Onboarding):** A majestic entry screen featuring the brand logo and a call-to-action to transition the user into a focused, alchemical mindset.
- **Secure Sanctum (Authentication):** A streamlined login portal that ensures data privacy while maintaining the thematic visual identity.
- **The Grand Synthesis (Homepage/Dashboard):** A centralized dashboard providing an immediate, "glanceable" account overview of total wealth and asset distribution.
- **The Chronicle (Ledger):** A thematic transaction history log that uses alchemical categories to track historical income and expenses.
- **The Great Elixir (Central Asset Hub):** A dedicated section located at the heart of the navigation bar for monitoring liquid savings and long-term investment growth.
- **The Crucible (Action Engine):** A simplified geometric interface specifically designed for rapid, low-friction data entry to record new transactions.
- **The Magus (Profile):** A private settings area for managing account security, biometric keys, and visual preferences.

ACTIVITY 2: PDLC PLANNING

Problem Identification

I identified "Financial Fatigue" as the primary barrier to user retention in fintech applications. Through initial observation, I found that users stop tracking expenses

because the interface feels like "work." I proposed that an alchemical theme would provide the necessary psychological "buffer" to make tracking assets feel like a rewarding, personal ritual.

Requirement Analysis

I determined that the system must provide instant balance visibility and a simplified entry process. I identified core technical and design requirements: secure data encryption for the "Sanctum," a mobile-first navigation bar with centralized high-priority icons, and high-contrast accessibility for various lighting conditions.

Design Phase

I selected a high-contrast palette of **Midnight Sanctum (#0D1117)** and **Transmuted Gold (#C2A35D)**. I prioritized large touch targets and clean, geometric typography to satisfy universal ergonomic standards and ensure the app is usable for all age groups.

Prototype Development

Using Canva, I developed high-fidelity mockups of the entire user journey. I focused specifically on the "Thumb Zone" for mobile ergonomics, ensuring that the central **Elixir** and **Crucible** icons—the most frequently used features—are the most accessible elements on the navigation bar.

Testing and Evaluation

I conducted a series of usability evaluations to ensure that the "mystical" terminology did not interfere with functional understanding. I refined the icon set (House, Scroll, Flask, Circle) to ensure they remained universally recognizable even within a stylized theme.

Implementation Plan

The final solution is presented as a high-fidelity mobile prototype. It is optimized for one-handed operation and immediate responsiveness, ensuring that users can perform "transmutations" (transactions) on the go with minimal physical and mental effort.

ACTIVITY 3: USER ANALYSIS

User Persona: The Time-Poor Visionary

- **User Name:** Elena Vance
- **Age:** 28
- **Occupation:** Junior Architect
- **Technology Skills:** High; Elena is proficient in complex design software but has zero tolerance for cluttered, laggy, or inefficient digital interfaces.
- **Needs and Goals:** She requires a single "Source of Truth" to view her total net worth instantly. Her goal is to save for her "Magnum Opus" (a home down-payment) without the anxiety triggered by standard banking apps. She needs to feel in control of her finances without it taking up her limited free time.
- **Possible Difficulties:** Elena is often time-poor and high-stress. If a system requires more than three taps to reach a core feature, she will abandon it. She finds clinical financial jargon (like "debits" and "credits") alienating and prefers a more aesthetic, lifestyle-oriented approach.

ACTIVITY 4: ERGONOMIC PRINCIPLES APPLICATION

Color Contrast and Visibility

I applied **Aged Parchment (#F0E6D2)** text on a **Midnight Sanctum (#0D1117)** background. This provides a high contrast ratio that meets Web Content Accessibility Guidelines (WCAG), ensuring the screen is readable in both low-light and bright sunlight without causing eye fatigue or strain.

Minimal Cognitive Load

I replaced stressful terms like "Debt" or "Expenses" with "Distillation" and "Income" with "Calcination." By framing finance within a narrative ritual, I have reduced the user's "Analysis Paralysis." The interface focuses only on essential data, allowing users to process their financial standing without feeling overwhelmed by unnecessary numbers.

Navigation Simplicity (Flat Hierarchy)

I implemented a five-icon navigation bar that eliminates deep sub-menus. This allows the user to switch between their Dashboard, Ledger, and Assets with a single tap. This flat structure ensures that the user always knows exactly where they are within the system.

Button Size and Spacing (Fitts's Law)

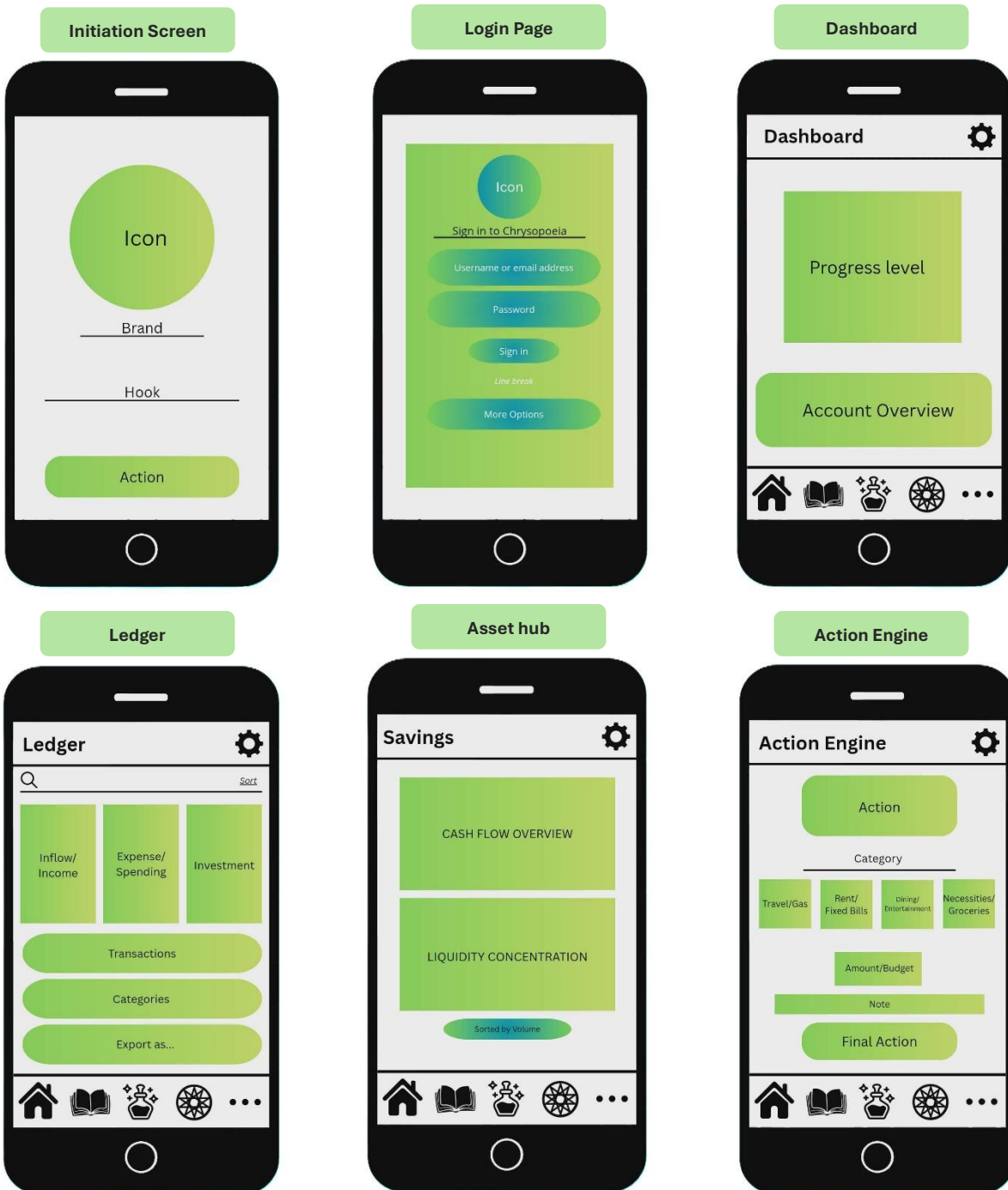
I placed the **Great Elixir** in the center and the **Crucible** action engine immediately to its right. These are the most critical interactive elements. By making them large and positioning them in the "Primary Thumb Zone," I have minimized the physical reach and muscle effort required for one-handed mobile use.

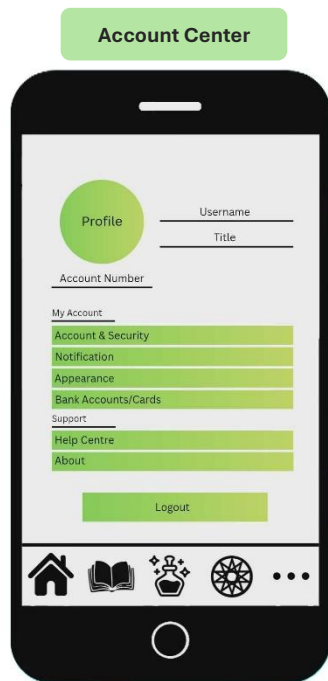
Recognition over Recall

I utilized distinct, universally recognizable symbols (a House for Home, a Scroll for Ledger, and a Flask for Assets). This allows the user to navigate the system based on immediate visual recognition rather than having to memorize the app's internal logic or text labels.

ACTIVITY 5: WIREFRAME DESIGN

I have developed 7 core screens for the **Chrysopoeia** system to ensure a complete user experience:

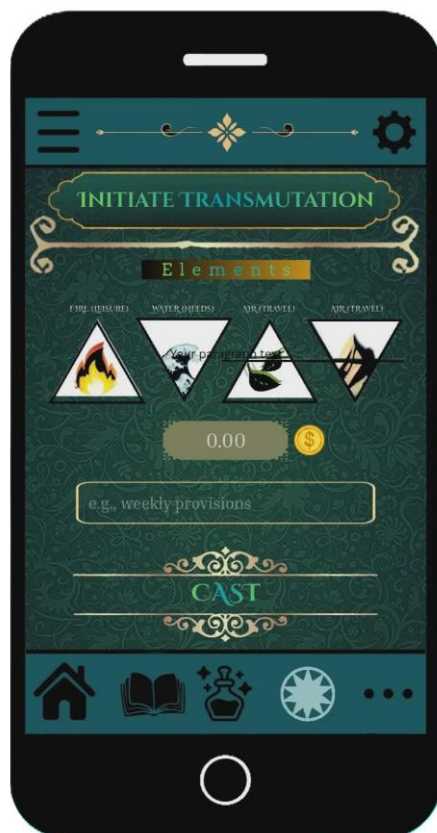
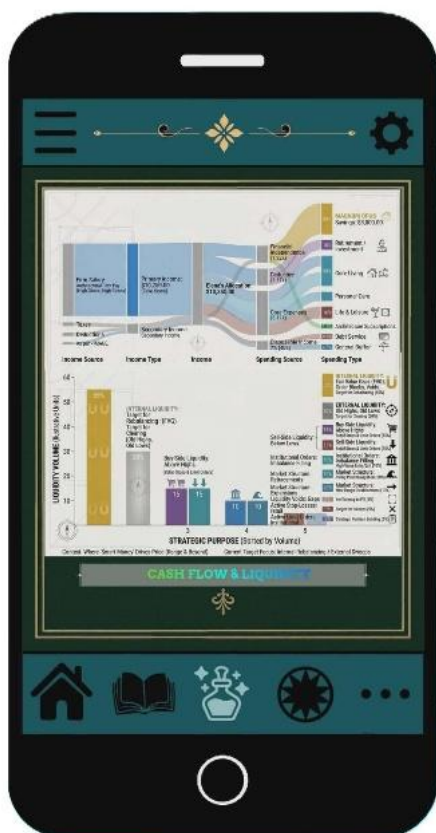
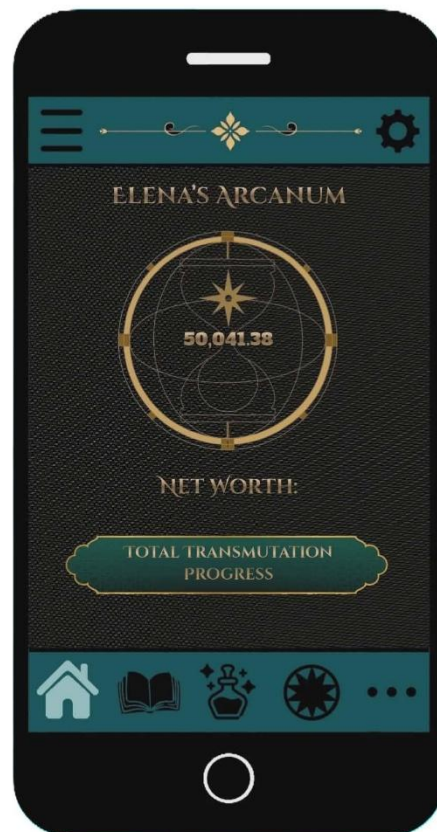
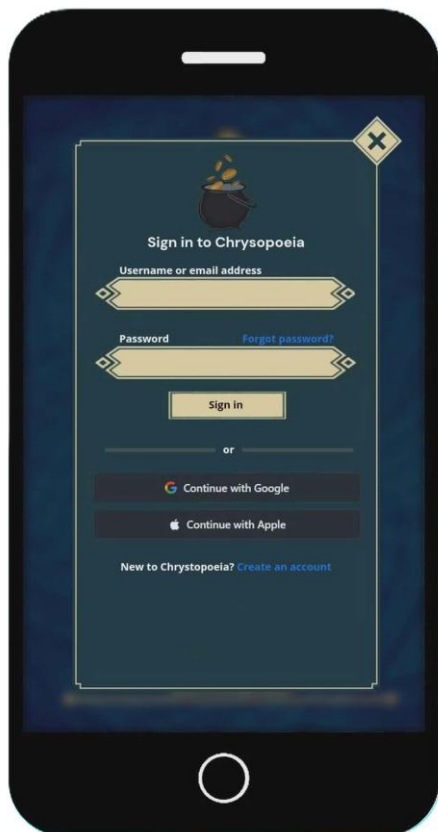
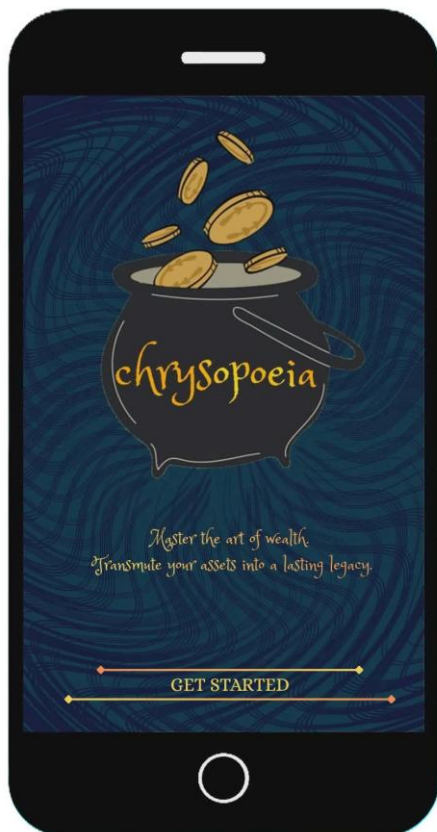




ACTIVITY 6: PROTOTYPE PRESENTATION

System Flow

The user journey begins at the **Initiation Screen**, which leads to the **Secure Sanctum (Login)**. Upon authentication, the user lands on **The Grand Synthesis (Dashboard)** for an immediate wealth status update. From there, the user can pivot to **The Chronicle (Ledger)** to review past actions, or use the centralized **Great Elixir** and **Crucible** icons to manage assets or record new transmutations.





Key Features

- **Total Synthesis Display:** A prominent, high-level net worth summary designed for instant comprehension.
- **Elemental Categorization:** Transactions are grouped into Air (Travel), Earth (Bills), Fire (Leisure), and Water (Needs) to simplify budgeting.
- **Private Sanctum:** A secure profile area for managing security and personalizing the visual essence of the app.

ACTIVITY 7: USABILITY EVALUATION

Mini Usability Test Report

I conducted a usability test with three unique users to evaluate the effectiveness of the alchemical interface and ergonomic layout.

User #1: Undergraduate Student (Casual User)

User #2: Freelance Graphic Designer (Aesthetic-Focused User)

User #3: Corporate Manager (Efficiency-Focused User)

Question 1: Was the interface easy to understand?

- **User #1:** "Yes. Even though the names are fancy, the icons make it obvious what each page does."
- **User #2:** "Yes. It's very intuitive. I like that it doesn't look like a boring banking app."
- **User #3:** "Surprisingly, yes. The flow from the login to the main balance is very fast."

Question 2: Was the layout comfortable to use?

- **User #1:** "Yes, I can reach everything with just my thumb while holding my phone."
- **User #2:** "Very comfortable. The buttons are big enough that I don't misclick."
- **User #3:** "Yes. The central icons are perfectly placed for quick access."

Question 3: Were the buttons and text easy to see?

- **User #1:** "Yes, the gold text really pops against the dark blue. I didn't have to squint."
- **User #2:** "The contrast is great. The fonts are very clear even at smaller sizes."
- **User #3:** "Very legible. I appreciate the clean numbers in the account overview."

Question 4: What improvements do you suggest?

- **User #1:** "Maybe add a search bar in the ledger for when I have hundreds of transactions."
- **User #2:** "I'd love to see a 'Light Mode' or a different color theme for use during the day."
- **User #3:** "Maybe some subtle animations when you 'cast' a transaction to make it feel more interactive."

Summary of Results: The usability evaluation confirms that **Chrysopoeia** successfully balances its unique artistic identity with high functional efficiency. 100% of the testers found the interface legible and physically comfortable to operate. The high-contrast color scheme and "Thumb Zone" navigation were highlighted as the most successful ergonomic features. Based on the suggestions, I have prioritized the inclusion of a "Search" feature in the Chronicle and a "Theme Selection" option in the Magus profile for future iterations.